

Objects in JavaScript – Deep Dive

Objects in JavaScript are used to store collections of **key-value pairs**. They are one of the most important and widely used data types in the language.

Creating an Object

You can create an object using **object literal syntax**:

```
let person = {  
  name: "Alice",  
  age: 30,  
  isEmployed: true  
};
```

Accessing Object Properties

You can access properties using **dot notation** or **bracket notation**:

```
console.log(person.name);    // "Alice"  
console.log(person["age"]);  // 30
```

Use bracket notation when the property name is stored in a variable or contains special characters:

```
let key = "isEmployed";  
console.log(person[key]);    // true
```

Modifying Object Properties

```
person.age = 31;  
person["name"] = "Bob";
```

Adding New Properties

```
person.city = "Delhi";  
person["hobby"] = "Reading";
```

Deleting Properties

```
delete person.hobby;
```

Checking if a Property Exists

```
console.log("age" in person); // true  
console.log(person.hasOwnProperty("city")); // true
```

Looping Through Object Properties

Use `for...in` to iterate over an object's keys:

```
for (let key in person) {  
  console.log(key + ": " + person[key]);  
}
```

Nested Objects

Objects can contain other objects:

```
let student = {  
  name: "John",  
  address: {  
    city: "Mumbai",  
    pin: 400001  
  }  
};  
  
console.log(student.address.city); // "Mumbai"
```

Objects and Functions

Objects can have **methods** (functions defined inside them):

```
let user = {  
  name: "Sara",  
  greet: function () {  
    console.log("Hello, " + this.name);  
  }  
};  
  
user.greet(); // "Hello, Sara"
```

You can also use shorthand syntax for methods:

```
let user = {  
  name: "Sara",  
  greet() {  
    console.log("Hello, " + this.name);  
  }  
};
```

Summary

- Objects store key-value pairs.
- Access and modify properties with dot or bracket notation.
- Use `for...in` to loop through properties.